Rising Chronic Kidney Disease in Bangladesh: A Hidden Crisis and a Proposal for Tech-Enabled Early Detection

Abstract

Background: Chronic Kidney Disease (CKD) is a growing but under-addressed public health concern in Bangladesh. From 1999 to 2023, CKD prevalence increased from 5% to 11%, alongside rising rates of diabetes (4.2% to 6.6%) and hypertension (12% to 15%). Despite this trend, access to non-communicable disease (NCD) screening remains limited (under 55%), and dialysis infrastructure remains inadequate. This study aims to highlight the increasing burden of CKD and propose a digital approach for early detection and awareness. Methods: An analysis of previously recorded public health data from 1999–2023 was conducted using secondary sources including WHO, BDHS, MOHFW, IHME, and the World Bank. The study analyzes trends in CKD risk factors, healthcare infrastructure, and government spending. Based on this analysis, a conceptual design is proposed for a mobile-based CKD screening app. The tool would use simple user inputs (age, blood pressure, diabetes status) and a rule-based logic system to assess risk and provide early warning prompts. Results: Findings show a steady rise in CKD prevalence with limited screening capacity and declining government health expenditure (from 7.5% to 5.1% of GDP). While the digital tool is in the planning stage, it presents a feasible, low-cost option for reaching at-risk populations in rural and low-resource settings. Conclusion: CKD is a silent but growing threat in Bangladesh. By incorporating digital technology into community-level screening, early detection can be significantly improved. This interdisciplinary approach aligns with national goals for combating NCDs and reducing healthcare burden.